

CSE331: Automata and Computability

Fall'25 | Assignment 1

Deadline: 13th November, 2025

Question 1 [10 marks]

Draw state diagram for a DFA of the following regular languages:

- A. $L = \{w \in \text{string that has "a" in the second last digit}\}, \Sigma = \{a, b\}$
- B. $L_2 = \{w \in \text{a string starts with 'ba' and length is odd}\}, \Sigma = \{a, b\}$
- C. $L_3 = \{w \in \text{a string where 1 is followed by at least two 0}\}, \Sigma = \{0, 1\}$
- D. $L_4 = \{w \in \{0, 1\} \mid w \text{ ends with 11 and does not contain the substring 10}\}$
- E. $L_5 = \{w \in \{a, b\} : \text{length of } w \text{ is multiple of 2} \cap \text{contains at least two a's}\}$

Question 2 [10 marks]

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$L1 = \{w \in w \text{ a string where after every 0's, 1 appears even times}\}$

$L2 = \{w \in \text{a string where 1 appears in every even position}\}$

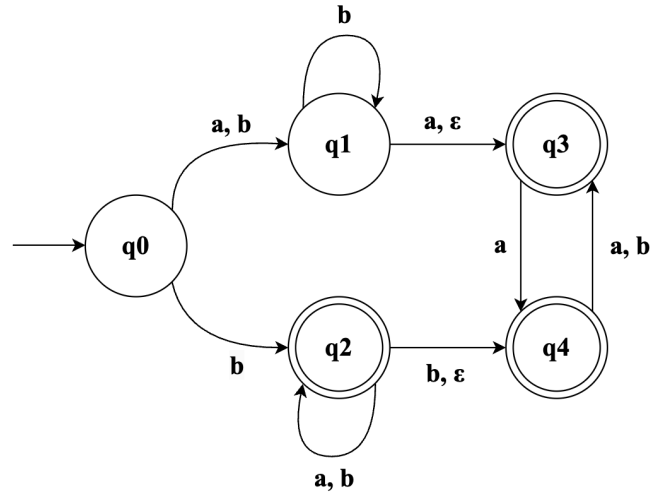
$L3 = \{w \in w \text{ a string that has subsequence of "01"}\}$

$L4 = L2 \cap L3$

$L5 = \{w \in \{10, 00\}^*\}$

- (a) Give the state diagram for a DFA that recognizes $L1$.
- (b) Give the state diagram for a DFA that recognizes $L2$.
- (c) Draw a DFA state diagram for $L3$.
- (d) How many states will be there for $L1 \cap L2 \cap L3$, if use cross product rule?
- (e) Draw a DFA state diagram for $L4$.
- (f) Find all the unique strings of **length four** for $L5^*$.

Question 3 [5 marks]



- (a) When converting the given NFA to an equivalent DFA using the subset construction method, what is the maximum number of states that the resulting DFA can have?
- (b) How many accepting states will be there while converting from the given NFA in its equivalent DFA?
- (c) Determine the ϵ -closure of state q_1 in the given NFA.
- (d) What is $\delta(\{q_1, q_4\}, b)$ in the given NFA? List all the states. [Recall: $\delta(\{q\}, 0)$ refers to the set of states the NFA transitions to when it is in state q and receives input a .]
- (e) What would be the start state of the converted DFA? Write the subset of the start state.

Question 4 [5 marks]

Draw state diagram for a NFA of the following regular languages:

- A. $L_1 = \{w \in a \text{ string that contains 'aba' } \cap \text{ ends with 'aab'}\}, \Sigma = \{a, b\}$
- B. $L_3 = \{w \in a \text{ string where third last symbol is 'b'}\}, \Sigma = \{a, b\}$

Question 5 [20 marks]

Let $\Sigma = \{0, 1\}$. Give regular expressions generating each of the following languages over Σ .

- (a) $\{w : w \text{ a string which has a subsequence of 'aba' } \cap \text{ starts and ends with different symbol}\}$
- (b) $\{w : w \text{ a string which length is not divisible by 2 } \cap \text{ w starts and ends with same symbol}\}$
- (c) $\{w : \text{every 1 in w is followed by an even number of 0s}\}$
- (d) $\{w : w \text{ a string where 0s and 1s are in alternate position } \cap \text{ number of 1s in w is divisible by 2}\}$
- (e) $\{w : 10 \text{ appears in w exactly once}\}$
(Hint: If $w = x10y$, what can you say about x and y?)
- (f) $\{w : w \text{ where the first symbol appears in every even position}\}$
- (g) $\{w : w \text{ containing strings where every third position in w is 1}\}$
- (h) $\{w : w \text{ containing strings where every 1 in w is followed by at least two 0}\}$
- (i) $\{w : w \text{ containing strings where every third position in w is 1 } \cap \text{ every 1 in w is followed by at least two 0}\}$
- (j) $\{w : w \text{ containing strings that end in three consecutive 1's}\}$

Question 6 [5 marks]

Convert Regular Expression to NFAs:

- (a) $(a^+b)^*b + a(ba)^*$
- (b) $((a^*b)c)^* + bb + a^+cb$

Question 7 [5 marks]

Convert the following DFA into an equivalent regular expression using the state elimination method. **Must follow this sequence to eliminate: B, D, C, F, E, A.** Show all steps.

